1 Project Description

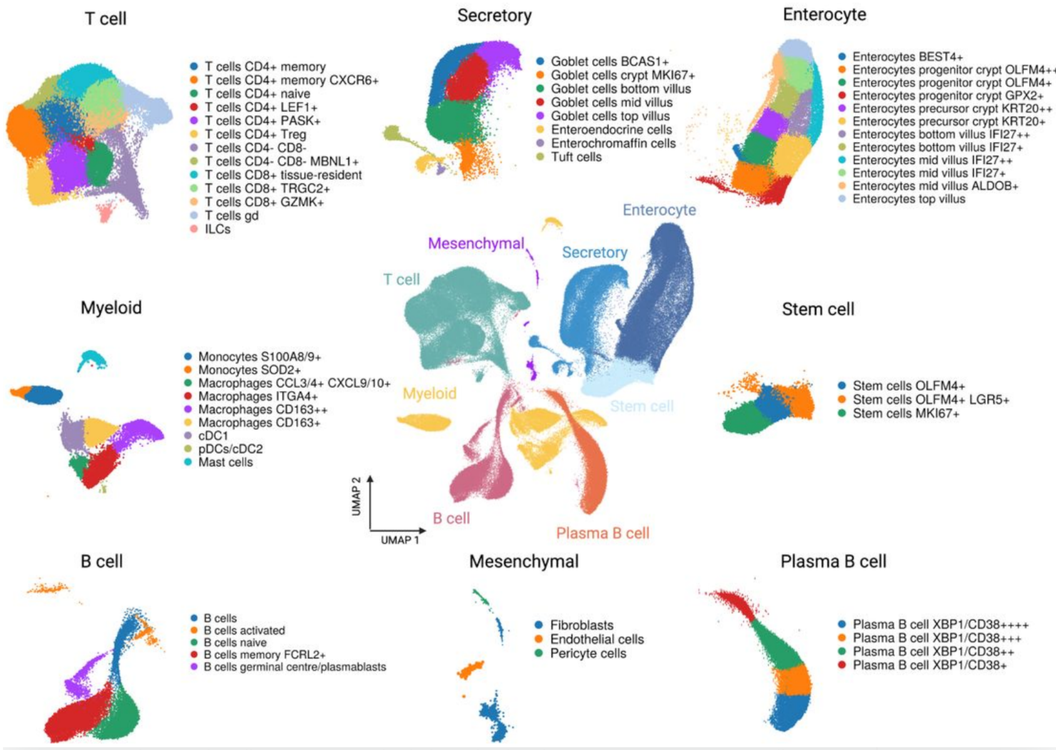

Figure 1: Application of UMAP to Single Cell RNA-seq data from the IBDVersedataset. The center figure displays the overall organization of the data, the peripheral figures display the organization of subsets. The colors correspond to types of cells (phenotypes)

1.1 Background

Large and high dimensional datasets are rapidly increasing in size, number and importance. For example, the Wellcome Sanger Institute recently published the IBDVerse dataset [9] for the study of Crohn's Disease (CD). The dataset contains RNA-sequencing signatures from some 1.1 million individual cells from the terminal ileum (a part of the small intestine) of 111 CD patients and 232 healthy individuals. This massive effort aims to identify the specific cells, genes, and biological pathways involved in CD.

The dimension of this data is the number of genes measured, which is typically around 20,000 for human cells. While all of the cells in the body of a single person are the same, the vectors measured by RNA-seq are specific for each cell and provide a detailed view of the cell's state, or phenotype. In fact, the view is so detailed, that it is very hard to compare between CD and healthy tissue using data in it's raw form. Dimensionality reduction is critical.

The dimensionality reduction methodology that is currently popular RNA-seq studies is to find a non-linear mapping of the sequences into a two dimensional plane in such a way that approximately preserves the neighborhood structure. Many methods for dimensionality reduction have been proposed over the years. However, the computation complexity of most of these methods makes them impractical for large datasets. For data of the size and dimension of IBDVerse the only viable method are currently attraction-repulsions force-based t-SNE [15] and UMAP [11].

Figure 1 displays the result of applying UMAP to the IBDVerse dataset. The colors in the figure indicate types and sub-types of cells. Comparing the figures generated from CD vs. healthy cells allows quantification of the relative plurality of different types of cells in different populations. The gene-based mapping is consistent with the main cell types. However, as is reported in [9] the mapping is less reliable regarding sub-types of Plasma B-cells.

This example demonstrates the tension between two goals. On the one hand, we need a dimensionality reduction method that scales to large datasets. On the other hand, we want a method that preserves the data structure at multiple levels of resolution. The goal of this proposal is to develop new algorithms that advance both goals.

RNA-seq is just one of many areas in which dimensionality reduction is of critical importance. Other areas include image and video data such as CIFAR-10, ImageNet, large-scale video datasets like Kinetics, text datasets such as large corpora, sensor and IoT data from smart devices, financial datasets like stock market indicators, astronomical datasets from surveys like SDSS, social network datasets such as those from Facebook or LinkedIn, and environmental data like climate simulations or satellite imagery.

Our proposal is to use a well-known dimensionality reduction methodology called diffusion maps, or eigen-maps, but use a different representation, we call local voltage maps, that allows for an efficient muti-resolution implementation.

1.1.1 Summary of Methods for non-linear dimensionality reduction

High-dimensional datasets often exhibit low intrinsic dimensionality, requiring methods that uncover the underlying low dimensional structure while preserving geometric and topological properties. Linear methods such as Principal Component Analysis (PCA) identify global axes of maximum variance [7]. Datasets with complex structure require non-linear methods. We group non-linear methods into three groups.

Attraction-repulsion force-based methods t-SNE [15] and UMAP [11] focus on preserving local relationships or geodesic distances on nonlinear manifolds. These methods excel in visualization and fast processing but may sacrifice some structural details in their reduced representations. All attraction-repulsion force-based methods can result in infinite distortion embeddings of the high-dimensional data depending on choice of a particular hyperparameter [10, 12]. Infinite distortion embeddings are problematic for downstream analysis, as they are either ripping apart otherwise connected data and to create the appearance of artificial clusters, or collapse far apart data points on top of each other in the embedding space to create the appearance that those points should be treated equivalently. These distortions have been demonstrated experimentally, especially when the data has a high aspect ratio or comes from a closed manifold [8].

Graph-based approach provide an alternative by constructing graphs where nodes represent data points and edges capture relationships such as proximity or similarity. Methods such as Isomap [13] characterize the structure of the graph using shortest distance paths. While spectral methods analyze the Laplacian of such graphs to identify global patterns [1].

Other non-linear frameworks include *topological data analysis (TDA)*, which captures multi-scale features such as connected components and loops [6], and *density-based clustering methods* like DBSCAN, which identify clusters based on density variations. While TDA emphasizes topological invariants and clustering methods focus on density, "Structure from Voltage" bridges these perspectives by using effective resistance to encode both geometric and topological complexity.

1.2 Concepts related to graphs

We briefly introduce some relevant concepts from graph theory.

1.2.1 The Graph Laplacian

The graph Laplacian is a fundamental construct in spectral graph theory, widely used to analyze the structure and properties of graphs. It plays a key role in applications such as dimensionality reduction, clustering, and network analysis.

A weighted undirected graph $G = (V, E, W)$ consists of a set of n vertices V, a set of undirected edges E, and a matrix of weights W associated with each pair of nodes. These weights represent the strength of the relationship between connected vertices. Figure **??** depicts a simple weighted graph whose **weight matrix** is

$$W = \begin{bmatrix} w_{00} & w_{01} & w_{02} & w_{03} & w_{04} \\ w_{10} & w_{11} & w_{12} & w_{13} & w_{14} \\ w_{20} & w_{21} & w_{22} & w_{23} & w_{24} \\ w_{30} & w_{31} & w_{32} & w_{33} & w_{34} \\ w_{40} & w_{41} & w_{42} & w_{43} & w_{44} \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 & 0 & 6 \\ 2 & 0 & 3 & 8 & 0 \\ 0 & 3 & 0 & 1 & 0 \\ 0 & 8 & 1 & 0 & 4 \\ 6 & 0 & 0 & 4 & 0 \end{bmatrix} \tag{1}$$

The **Laplacian matrix** L for a graph is defined as:

$$L = D - W$$

where D is the **degree matrix**, a diagonal matrix where each diagonal entry d_{ii} is the sum of the weights of the edges connected to vertex i, i.e., $d_{ii} = \sum_j w_{ij}$. In terms of elements, the Laplacian matrix L can be expressed as:

$$L_{ij} = \begin{cases} d_{ii} = \sum_j w_{ij}, & \text{if } i = j \\ -w_{ij}, & \text{if } i \neq j \text{ and there is an edge between } i \text{ and } j \\ 0, & \text{if } i \neq j \text{ and there is no edge between } i \text{ and } j \end{cases}$$

The Laplacian matrix corresponding to the weighted matrix W defined in Equation (1) is

$$L = \begin{bmatrix} 8 & -2 & 0 & 0 & -6 \\ -2 & 13 & -3 & -8 & 0 \\ 0 & -3 & 4 & -1 & 0 \\ 0 & -8 & -1 & 13 & -4 \\ -6 & 0 & 0 & -4 & 10 \end{bmatrix}$$

Having no edge between i and j is equivalent to setting $w_{ij} = 0$. We thus assume without loss of generality that the graph is complete, i.e., there is an edge between any two vertices, with weights of zero where edges are absent.

There are two main ways to characterize the Laplacian matrix L: the spectrum and the minimization of a quadratic form.

- **The Spectrum:** The spectrum of the matrix L (or the spectrum of the graph) is the set of eigenvalues and corresponding eigenvectors of L, ordered by increasing eigenvalue. The dominant eigenvectors, associated with the smallest eigenvalues, are typically smooth, global functions that capture the overarching structure of the graph. These eigenvectors are widely used in tasks such as clustering and dimensionality reduction.

- **Minimization of a Quadratic Form:** Another way to characterize the matrix L is through a quadratic form. Given a vector $x \in \mathbb{R}^n$, the quadratic form of the graph Laplacian is defined as:
$$E(x) = x^T L x = \sum_{(i,j)\in E} w_{ij}(x_i - x_j)^2, \tag{2}$$
$E(x)$ is sometimes called the *energy* of the configuration x. Of particular interests are configurations that minimize the energy under a set of constraints. Considering the energy minimizing configurations under different constraints provides insights into the properties of the weighted graph.

In summary, the graph Laplacian serves as a powerful mathematical tool to uncover both local and global properties of a graph. Next we consider a specialized kind of graph that captures the structure of cloud points.

1.2.2 Random Walks

The graph Laplacian can be used to model random walks on the graph G. Imagine a large number of ants distributed among the vertices of the graph. At each time step, each ant moves to one of the neighboring vertices at random, with probabilities defined by the transition matrix L. The eigenvectors of the Laplacian capture the stationary distribution and the rate of convergence to it, with the smallest non-zero eigenvalue providing insights into the graph's connectivity and mixing time.

Eigenvectors are not the only way to characterize the matrix L. An alternative characterization is provided by the *commute time* between two vertices u and v, defined as the expected time it takes for an ant starting at u to reach v and then return to u (or vice versa). The commute time between pairs of nodes defines a metric between the vertices of the graph that characterizes the graph at both the local and global levels, depending on the relative locations of u and v.

This more local characterization is mathematically identical to the concept of effective resistance, which will be described next.

1.2.3 Effective Resistance

Another perspective on the graph Laplacian is as a model of resistor circuits [5]. In this case, the graph represents an electric network where the vertices are connection points, and the edges are resistors. The *effective resistance* between two vertices u and v can be calculated by setting the voltage at u to 1 volt, the voltage at v to 0, the minimize the Laplacian configuration energy to compute the voltage at all other vertices. The effective resistance is given by $1/I$, where I is the total current between u and v. The effective resistance between two vertices can also be expressed in terms of the pseudo-inverse of the Laplacian.

As it turns out, the effective resistance is mathematically equivalent to the commute distance between u and v.

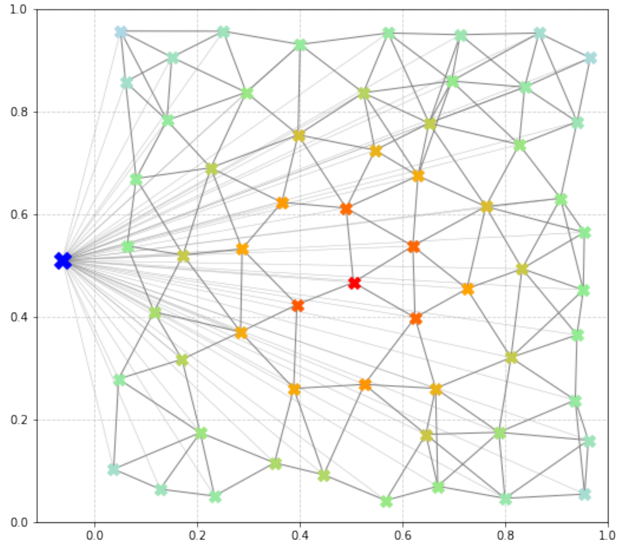

Figure 2: A schematic of a grounded resistor graph and a local voltage map.

1.2.4 Grounded resistance Graphs

In [3] we introduce a novel type of resistor graph we call the grounded resistor graph(GRG)(Fig 2. The reason GRGs are useful for mapping will be explained in Section 1.5.

To make any resistor graph into a grounded graph we add a node called "ground" (marked by a blue X in the figure). This ground is connected to each of the nodes in the original graph by a large resistor whose conductance $\rho = 1/R$ is close to zero.

We associate a *local voltage maps* (LVM) with special vertices we call *landmarks*. An LVM is defined by setting the voltage of the landmark to 1 and the voltage of the ground to zero. The LVM is defined by setting the unconstrained nodes so that the Laplacian total energy is minimized.

We say that LVMs are "local" because the voltage decays exponentially to zero as the distance from the source increases. (figure 2)

We define a divergence $D(v, x)$ between a landmark v and a vertex x as one minus the voltage induced at the vertex x by the LVM centered at u.

The divergence $D(v, x)$ defined this way is an inverted peak which is zero at v and increases with the geodesic distance from v. Moreover the shape of the divergence function is upper and lower bounded by functions of the form

$$1 - \exp(-d(v, x)f_1(\rho) \leq D(v, x) \leq 1 - \exp(-d(v, x)f_2(\rho)$$

where $d(v, x)$ is the geodesic distance between x and v, ρ is the conductance to the ground and f_1, f_2 are increasing functions. In other words, the rate of decay is inversely proportional to the ground conductance ρ. This means that by controlling ρ we control the "width" of the voltage function. This is an important property that allows us to decouple the calculations of the LVM's for landmarks that are distant from each other, and solve the LVMs in parallel.

1.3 Graph-based analysis of point Clouds

The concepts introduced in Section 1.2 apply to any (weighted) graph. Here we are interested in applying them to graphs that represent the internal structure of large point clouds. More formally,

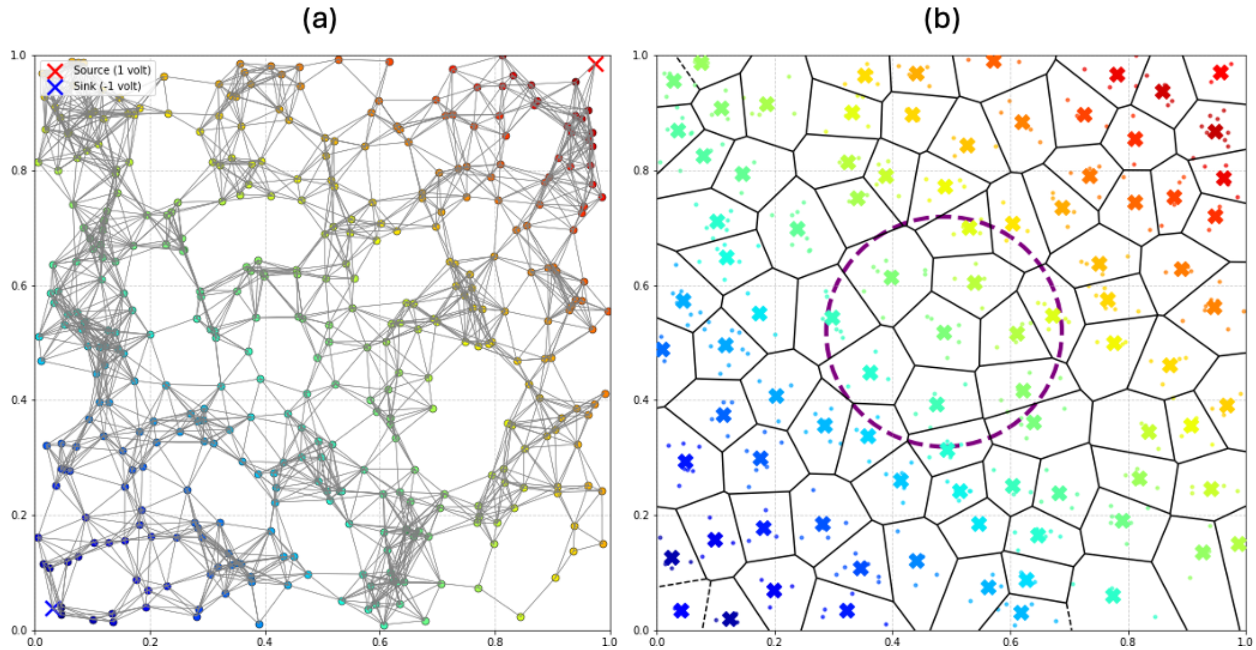

Figure 3: Two ways of computing a voltage map. In both figures the upper right corner is at voltage $V = +1$ (red) and the lower left corner is at voltage $V = -1$ (blue). Both figure use the disk kernel. **(a)** using the graph induced by the sample points, **(b)** using the graph induced by a K-means.

we assume the existence of a probability distribution from which data points are sampled IID. Our goal is to design a smooth mapping.

We wish to define a series of weighted graphs based on these points such that the Laplacians of this series of graphs has a well-defined limit that captures the structure of the distribution.

We present two alternatives to defining the vertices of the graph. The standard definition, which we call "point-based" and a new definition, which is alternative definition which we call "partition based". We describe each of these definitions and compare them, but first, we define what is common to both.

Given a set of vertices, that are points in the ambient space X, we define the weight of the edge connecting two any two vertices $x, y \in X$ as a function of the distance between x, y the so-called *Kernel* of the Laplacian. Two popular Kernels are the Gaussian and the Disk:

$$K_{Gauss}(x,y) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{\|x-y\|_2^2}{\sigma^2}\right); \quad K_{Disk}(x,y) = \begin{cases} \frac{1}{\pi R^2}, & \text{if } \|x-y\| \le R, \\ 0, & \text{if } \|x-y\| > R, \end{cases}$$

Th Gaussian kernel is easier to analyze, while the disk kernel has the advantage that it's support is bounded.

The common way to define the limit is point based, however, that approach suffers from several limitations. We propose an alternative way of taking the limit, which we call "partition based". We describe these approaches below.

Point Based In the point based approach each data point corresponds to a vertex of the graph. See Figure 3**a**.

Partition based We propose an approach to computing the local voltage maps whose atomic element is a cell, rather than a point. See Figure 3**(b)**. In this approach, the space is partitioned using K-means. The result is a partition into cells $i = 1, \ldots, n$. The mean of cell i defines a graph vertex $v(i)$ and a probability $p(i)$, corresponding to the fraction of datapoints in the cell.

Vertices $v(i), v(j)$ are connected by an edge if the distance between them is at most r. The weight of the edge is $p(i)p(j)$. Energy minimization is used to define the voltage function.

The partition-based solution has several advantages over the point-based solution in terms of computation:

- computation required to minimize the energy depends on the number of cells, rather than the number of points.
- Unlike a grid based partition whose size depends on the the ambient dimension, the number of cells generated by k-means, in turn, depends on the intrinsic dimension of the data, which is assumed to be much smaller.
- The two operations that require processing of data points are the generation of the partition and the estimation of the probabilities $p(i)$. These operations can be done using streaming, processing small batches of points and then discarding them.

Analysis In [2] Belkin and Niyogi established that, if the distribution is uniform on a compact Riemannian manifold, then the graph Laplacian operator converges to the Laplace-Beltrami operator on the underlying manifold as sample size increases and graph neighborhood size approaches zero at specific rates. Their proof showed that the discrete graph Laplacian constructed from randomly sampled data points serves as a consistent estimator of the geometric structure represented by the Laplace-Beltrami operator. This result provided mathematical justification for graph Laplacian methods in dimensionality reduction and data analysis by connecting discrete computational approaches to continuous geometric theory.

In [**?**], Hein et al extended these convergence results by analyzing random neighborhood graphs under weaker assumptions about the underlying space. The authors proved convergence of graph Laplacians for data sampled from probability distributions with support on arbitrary subsets of Euclidean space, rather than requiring the data to lie on a smooth manifold. Their work established pointwise convergence of the normalized graph Laplacian and derived explicit convergence rates in terms of the neighborhood size and sampling density. The paper also demonstrated that different graph Laplacian constructions converge to different limit operators, providing guidance for choosing appropriate normalizations based on the intended application.

Despite the convergence of the Laplacian, [16] shows that the point based approach does not extend to the effective resistance. This paper demonstrated fundamental limitations in the resistance distance metric for random geometric graphs. Their mathematical analysis proved that in large random geometric graphs under certain normalizations, the resistance distance between most vertex pairs converges to a constant value that depends primarily on the vertices' degrees rather than their meaningful geometric relationships. The authors established this convergence behavior for both k-nearest neighbor and ε-neighborhood graphs under standard scaling conditions of the connectivity parameters, effectively demonstrating that resistance distance becomes uninformative precisely in the asymptotic regime where many graph-based algorithms are meant to operate.

we show that this result on the triviality of effective resistance can fortunately be sidestepped under different normalizations of the source and sink [**?**], and, in particular, when we introduce grounded resistance graphs [3].

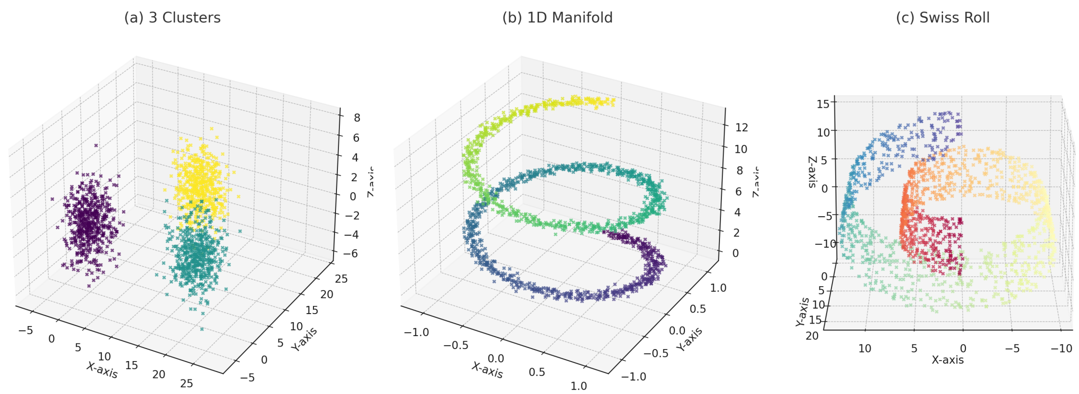

Figure 4: **An illustration of intrinsic dimension:** The point clouds in 3D that can be approximated by low dimensional manifolds. **(a)** three zero-dimensional clusters can be approximated by the three centers of mass. **(b)** a one dimensional spiral. **(c)** a two dimensional swiss roll.

1.4 Non-Uniform Local Dimension

So far, we discussed the application of the graph Laplacian to point clouds. It is time to talk about dimensionality reduction. Clearly, one cannot hope to achieve meaningful dimensionality reduction for an arbitrary dataset. For example, one cannot reduce the dimension of a dataset that is pure noise. Some assumptions are needed.

A common assumption is that the data has low *intrinsic dimension* or lie close to a low-dimensional manifold. Figure 4 contains examples of data with intrinsic dimensions of zero, one, and two, embedded in three-dimensional space. If the intrinsic dimension of a dataset is at most two, it is, in principle, possible to construct a low distortion, low dimensional mapping.

Describing an entire dataset as having a single intrinsic dimension (e.g. 0, 1, 2, or k) is overly simplistic. The intrinsic dimension is not a global property but can vary with location. For example, consider parts (a), (b), and (c) of Figure 4 residing at different locations in the same space; in such a case, the intrinsic dimension varies between different locations. Local dimension can also depend on on scale, consider the shape of a tree. The tree as a whole is a 3D object, each leaf is a 2D object, and branches are 1D objects.

Most dimensionality reduction methods fail to account for variability in intrinsic dimension with location and scale.

A major aim of this proposal is to overcome this limitation with *mapping using landmarks* as explained in the following section.

1.5 Localization using Landmarks

In this section we describe an application of the concepts described above to the problem of dimensionality reduction. **This is a description of our prior work on the subject** [**?**].

The core idea of our method is simple. Our input is a graph that represents a point cloud. We designate a set of n vertices in this graph as "landmarks" $\vec{u} = (u_1, \ldots, u_n)$. We also have a divergence $D(u, v)$ which quantifies the distance between each landmark and each other vertex. [1]

Given the set of landmarks $\vec{u}$ we parametrize an arbitrary vertex v by the vector of divergences $d(u_1, v), \ldots, d(u_n, v)$. This is akin to localization using GPS, where the distances to satellites

[1] We call this measure a divergence rather than a distance because it might not be symmetric or obey the triangle inequality.

(landmarks) are used to infer the coordinates of a cell phone.

If the graph lies on a manifold of dimension d the number of landmarks that are needed to disambiguate any pair of vertices is $d+1$. We conjecture that if the *local dimension* of the data in some region is d then the number of landmarks required is $O(d)$. When a dataset has different local dimensions in different locals one can use the union of the landmarks. The result is a dimensionality reduction method that adapts to local dimension.

We describe two variants of our approach, one based on effective resistance (ER) (Section 1.2.3) and the other based on LVMs (Section 1.2.4).

Landmarks based on ER ER defines a *metric* over the point cloud which captures the distribution of the cloud point as described in Section 1.2.3.

The fact that ER is a metric (symmetric and obeys the triangle inequality) it is easier to design the landmarks. Computing the pairwise divergences between landmarks and vertices requires storing all of the graph in memory. This cost becomes prohibitive for large graphs. Which leads us to the LVM approach.

Landmarks based on LVMs Our approach for scaling the landmark method to very large graphs is a divide and conquer approach. We partition each graph to partially overlapping regions, one region per landmark, and compute the divergence from the landmark to all other vertices. Performed naively, this would still require storing the full graphs. However a property of LVM divergences, which is not shared with RE divergences, is that they converge to 1, as the geodesic distance increases to infinity. The *rate* of convergence is controlled by ρ, the conductance to the ground.

As a result, the divergence between a landmark and a far away vertex can be implicitly approximated by 1 and not stored. When ρ is large only a small set of vertices that are close to the landmark need to be stored in memory in order to compute the divergences. On the flip side, if this set is small there is little overlap between the sets and triangulation will not work. The solution is to choose ρ to make the sets small but still keep a significant overlap betwen the sets of neear landmarks.

To summarize, our method of LVM based triangulations has the following advantages.:

- **Locality of mapping:** The divergence from far away landmarks is very close to 1 and is therefor not informative. In other words, the conductance can be used to control the range of imfluence of the landmarks.
- **limiting computation**: The computation of the minimal energy solution can be terminated when the values get close enough to 1.
- **distributed computation:** The computation of the voltage function for different landmarks can proceed in parallel. Each process storing only the local neighborhood of the landmarks.

1.6 Outline of algorithm

To illustrate how the different components come together We sketch our proposed algorthm:

1. The dataset is randomly permuted so that any prefix read as a stream can be seen as a random sample.
2. The algorithm operates top down, starting with a rough partition of the whole data-set and proceeding to increasingly finer and smaller partitions that cover the space with some overlaps.

3. The partition at each level is generated using streaming k-means++ followed by streaming k-means.

4. In each partition a set of landmarks and a conductance for each landmark is chosen so that the divergences between the landmarks and each vertex disambiguate each vertex.

5. Computation of the LVM for each landmark is done independently using in parallel using only the part of the graph that is close to the landmark.

6. The computation of LVM at each level of resolution is initialized with the solution at the previous level.

Given a multi-resolution embe for a dataset we can create 2D visualizations by selecting a small number of landmarks at a particular resolution and mapping the partition vertices at the given resolution to the plane using a standard non-linear mapper.

2 Objectives and Research Questions

3 Research Plan

The goal of this project is to develop a dimensionality reduction method with the following properties:

- Reliably represent the neighborhood structure at multiple scales.
- Scale efficiently to large datasets of high dimension.
- Allow easy visualization of parts of the data.

Our plan is to use multi-resolution landmarks and voltage maps to achieve these goals. The research questions that face us are of two types:

1. **algorithmic** Develop a streaming distributed implementation of data mapping using local voltage maps. Evaluate the performance of this approach on real-world datasets.

2. **Theoretical** Develop a mathematical theory for mapping data with large extrinsic and low intrinsic dimension.

Algoritmic questions The design of our algorithm is based on the following objects: data streams, partitions, landmarks, voltage functions and resolution levels.

The main algorithmic challenges we will address are:

1. **space partitioning** How to partition the input space into a small number of cells so that the graph Laplacian base on those cells is an accurate representation of the data distribution. Methods under consideration include K-means++ [**?**] followed by K-means to convergence, cover trees [**?**] and ϵ cover trees[**?**].

2. **Landmarks selection** How to select a small number of landmarks that create a bi-Liphsitz mapping between the data space and the triangulation space. Initial methods under consideration are random choice and greedy selection.

3. **Parallelization** What hierarchical data structures will allow parallelization of the construction of the voltage maps and of the localization by triangulation.

   Our planned approach is to control both the width of each voltage map and the number of parts in the space partition. The width of the voltage maps will be set so that the overlap between voltage maps for different landmarks overlap enough to allow accurate triangulation yet narrow enough that each voltage map can be solved using a small fraction of the space. This will be achieved by increasing the resolution of the space partition at each level and setting the ground resistance for each landmark separately. The result would be that solving the energy minimization task for each landmark will require constant time and space. These tasks can be performed in parallel in a distributed manner.

4. **Comparative study** Run experiments to compare our approach to state-of-the-art approaches such as Laplacian Eigen-maps and t-SNE. We will use cluster quality metrics and KNN-classification accuracy, among other mesures, to quantify the consistency between a low dimensional mapping and independent acquired labels of the instances, such as cell-type in the case of RNA-seq.

Theoretical Questions The main research questions for theoretical analysis are:

1. **Limit properties of the voltage function** Prove that grounded voltage functions converge to non-trivial solutions as the graph size grows. We also seek to determine the correct scaling laws for the the partition diameter, the kernel radius, and the resistance to the ground to converge to non-trivial solutions. Our planned approach is to show that the voltage curves converge to the Green's function of a nonlocal Laplace-Beltrami operator. Similarly, we seek to extend preliminary results [**?**], which show the voltage curve has exponential isotropic decay on the sphere, to general data domains (e.g. manifolds, data that is locally low dimensional). For example, on Riemannian manifolds the desired result would be to show the effective support of the voltage curve forms a geodesically convex region whose radius grows as the grounding resistance grows.

2. **Convergence properties of Partition based Laplacian** Prove that the partition based framework described above converges to the nonlocal Laplace-Beltrami operator as the number of parts and the go to infinity at the appropriate rates. Our planned approach is an augmentation of [**?**], which considers the convergence of the affinity kernel between points using landmarks to define the affinity, and [**?**], which considers convergence of a different manifold learning scheme using ε-cover trees. While neither method solves our question, techniques can be borrowed to establish convergence of our partition based scheme for effetive resistance.

3. **Bi-Lipschitz embeddings** Prove that Euclidean distances in the embedding space of a partition at a certain resolution is bi-Lipschitz with respect to the natural distance in the original data space at that resolution. This will guarantee that the embedding has finite distortion and will not yield phantom clusters or collapse points. We note that we only seek such guarantees to some non-zero scale that is above the level of noise in the data.

4. **Local dimension at a given resolution** Derive meaningful definitions of dimensionality of data at a given resolution and location that go beyond the traditional notions of manifold dimension. Our planned approach revolves around the rate of diffusion of a random walk with finite variance, as there is an equivalence between certain diffused random walks and effective resistance as shown in [**?**]. An alternative approach would be dimensionality defined by the number of landmarks required to specify the location of a point up to some resolution.

3.1 Preliminary Results

- **Theoretical Guarantees:** We proved the existence and convergence of grounded voltage solutions, overcoming the limitations of traditional spectral methods.
- **Synthetic Experiments:** Experiments on a sphere show that voltage solutions decay exponentially away from the source, capturing local structure effectively.
- **Real-World Validation:** Applied to MNIST and Frey-face datasets, the method demonstrated robust performance and efficient computation.

3.2 Broader Impacts

The fruits of the proposed research are likely to have significant broader impact. Currently, tools that can handle large datasets generate visualizations often significantly distort the data while tools that do not distort the data cannot handle large datasets.

Areas of impact include

- Applications in machine learning, bioinformatics, and scientific computing for large-scale, high-dimensional data.
- Enabling distributed computation for dimensionality reduction, beneficial for resource-constrained systems.
- Open-source tools and datasets to promote accessibility and reproducibility.

3.3 Timeline of Work

- **Year 1:** Development of mult-resolution algorithm. Theoretical analysis of convergence and shape properties of grounded voltage solutions.
- **Year 2:** Development of scalable algorithms for distributed computation. Development of theory and algorithms for placing landmarks.
- **Year 3:** Experimental evaluation on synthetic and real-world datasets.